This tool is responsible for the data transformation in our project

Azure Data Lake Storage is an independent resource (not automatically linked to Databricks).

To access data from Azure Databricks, you need:

Permissions

Credentials

- How access works

You create a Service Application (Service Principal) in Azure

This acts as a secure identity / credential

This Service Principal is:

Given access (roles/permissions) to Data Lake

Used by Databricks to authenticate

- Flow

Create Service Principal (App)

Grant it access to Data Lake

Use its credentials in Databricks

Databricks requests access → shows credentials → gets data

- Analogy

Databricks = user

Service Principal = ID card/ticket

Data Lake = secured place

Databricks manages clusters using VMs inside a managed resource group—you just use it, not maintain it

Create a new resource databrics

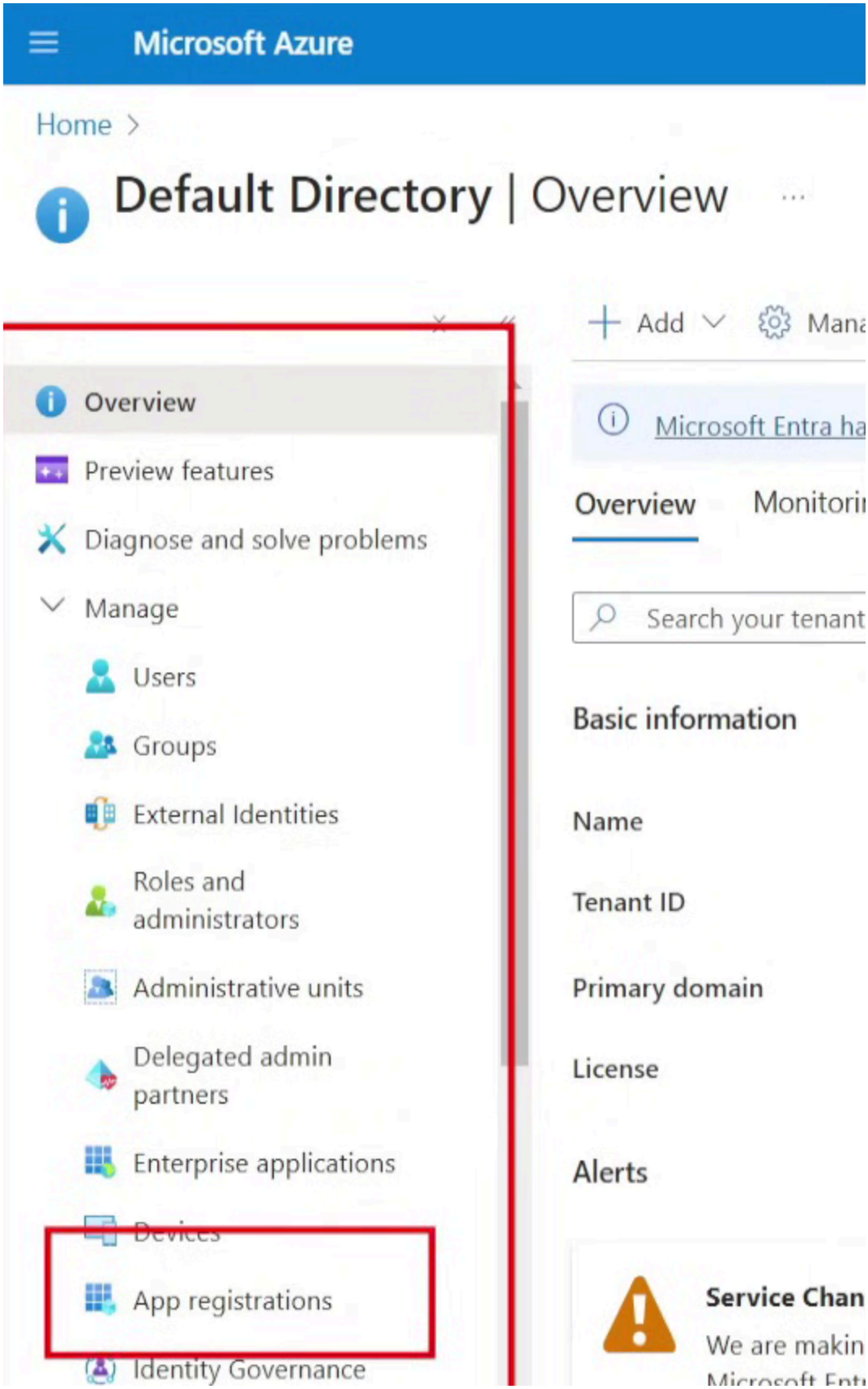
We will use compute to transform and process our data.

It's just a cluster

Create a new compute

Search Microsoft Entra ID

Go to app registrations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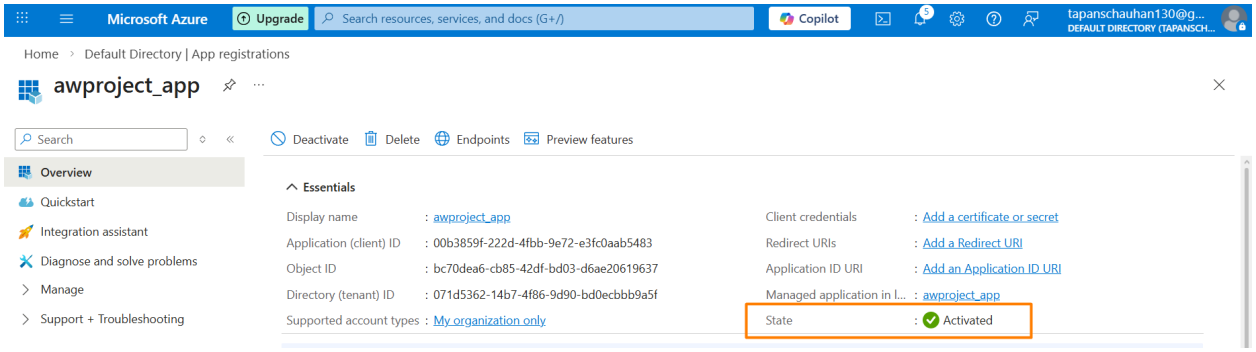


Now we need to assign this application a role which can contribute to our data lake

Save the above information as it will be needed afterwards

Why we will be pulling this application in databrics we need to pass the URL

Save application ID and object ID and now create secret to create secret click on certificate and secrets

```
app_id = "00b3859f-222d-4fbb-9e72-e3fc0aab5483"
object_id ="bc70dea6-cb85-42df-bd03-d6ae20619637"
secret = "Oc68Q~wBQiOScHIm7U7Jy1dRYQtkGG9c~sQPJcz-"
secret2 = "28c19c1e-4eff-443c-ae90-410da2e97e3d"
```

Important copy the secret value not the ID

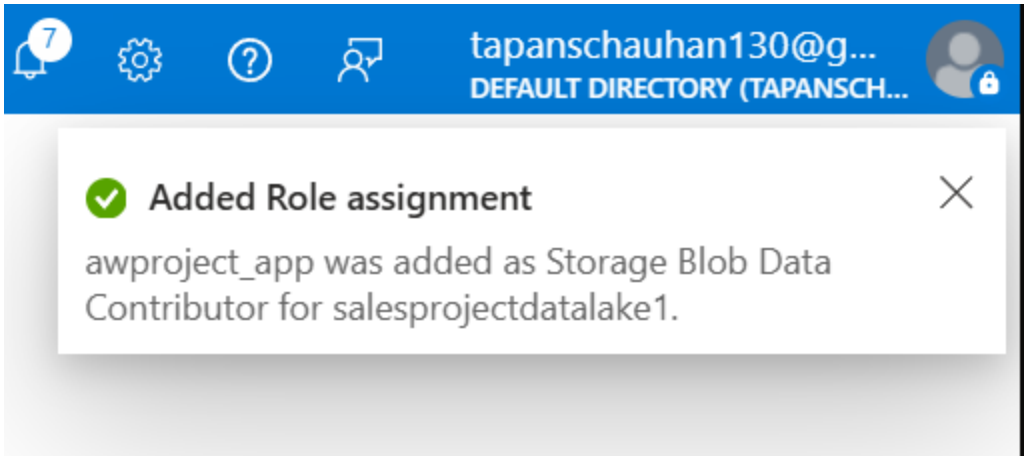
Step 2

Assign the role to the application

Go to your storage account

ADD IAM THE ROLE THAT WE WANT TO APPLY IS STORAGE BLOB CONTRIBUTOR IT CAN READ IT CAN WRITE

Select members add your aw project



Step 3

Open databrics

Check your created cluster is turned on or turn off

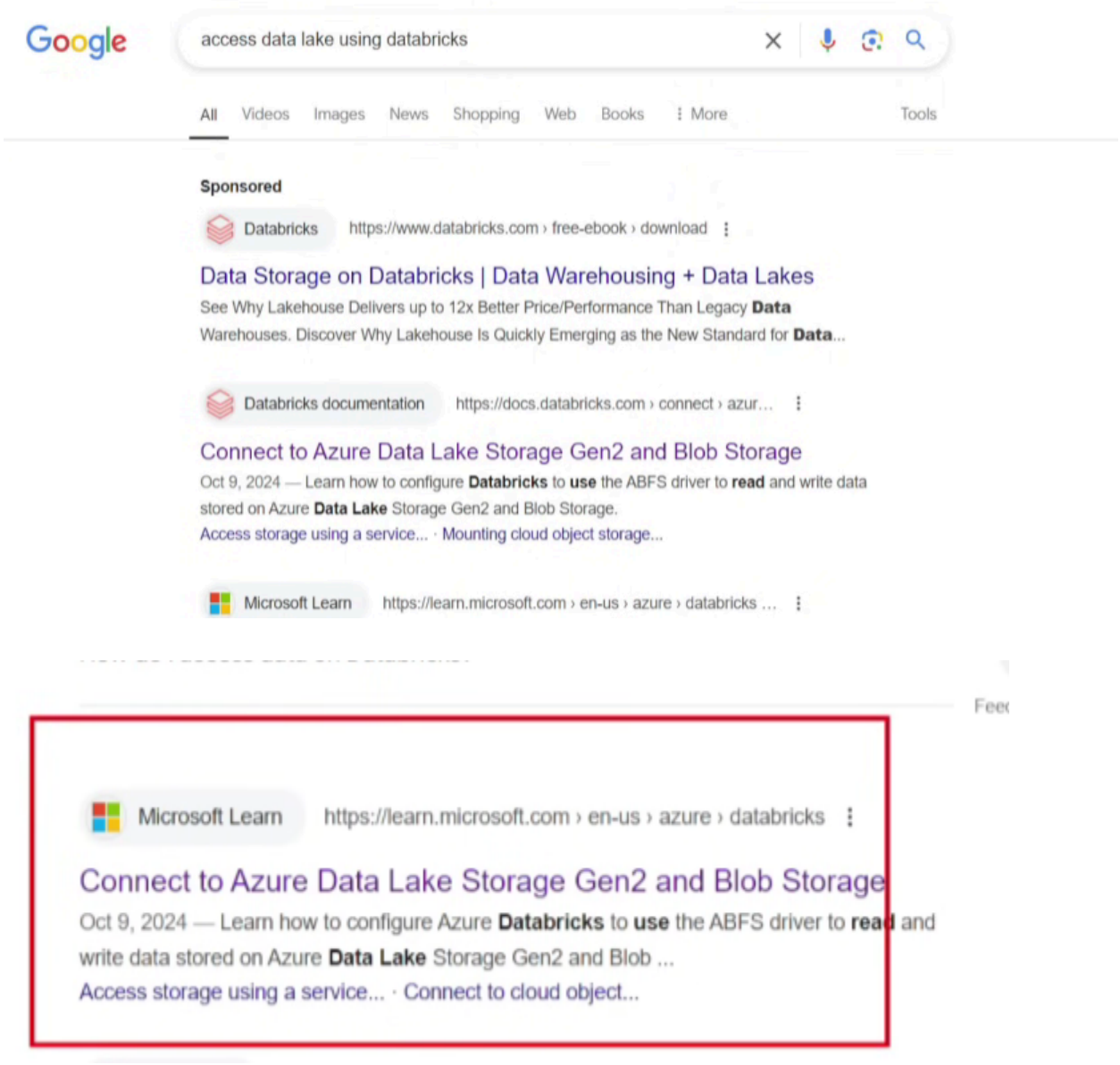
Click on connect and turn it on

Click on work space create a new folder now inside this folder create a notebook

On markdown write silver layer script

To run press shift plus enter

Get the documentations from Google



https://docs.azure.cn/en-us/databricks/connect/storage/azure-storage

Azure service principal

Use the following format to set the cluster Spark configuration:

```ini
spark.hadoop.fs.azure.account.auth.type.<storage-account>.dfs.core.windows.net OAuth
spark.hadoop.fs.azure.account.oauth.provider.type.<storage-account>.dfs.core.windows.net o
spark.hadoop.fs.azure.account.oauth2.client.id.<storage-account>.dfs.core.windows.net <app
spark.hadoop.fs.azure.account.oauth2.client.secret.<storage-account>.dfs.core.windows.net
spark.hadoop.fs.azure.account.oauth2.client.endpoint.<storage-account>.dfs.core.windows.ne
```

You can use `spark.conf.set` in notebooks, as shown in the following example:

```python
service_credential = dbutils.secrets.get(scope="<secret-scope>",key="<service-credential-k

spark.conf.set("fs.azure.account.auth.type.<storage-account>.dfs.core.windows.net", "OAuth
spark.conf.set("fs.azure.account.oauth.provider.type.<storage-account>.dfs.core.windows.ne
spark.conf.set("fs.azure.account.oauth2.client.id.<storage-account>.dfs.core.windows.net",
spark.conf.set("fs.azure.account.oauth2.client.secret.<storage-account>.dfs.core.windows.n
spark.conf.set("fs.azure.account.oauth2.client.endpoint.<storage-account>.dfs.core.windows
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spark.conf.set("fs.azure.account.oauth.provider.type.<storage-account>.dfs.core.windows.ne
spark.conf.set("fs.azure.account.oauth2.client.id.<storage-account>.dfs.core.windows.net",
spark.conf.set("fs.azure.account.oauth2.client.secret.<storage-account>.dfs.core.windows.n
spark.conf.set("fs.azure.account.oauth2.client.endpoint.<storage-account>.dfs.core.windows
```

Make sure to select windows.net

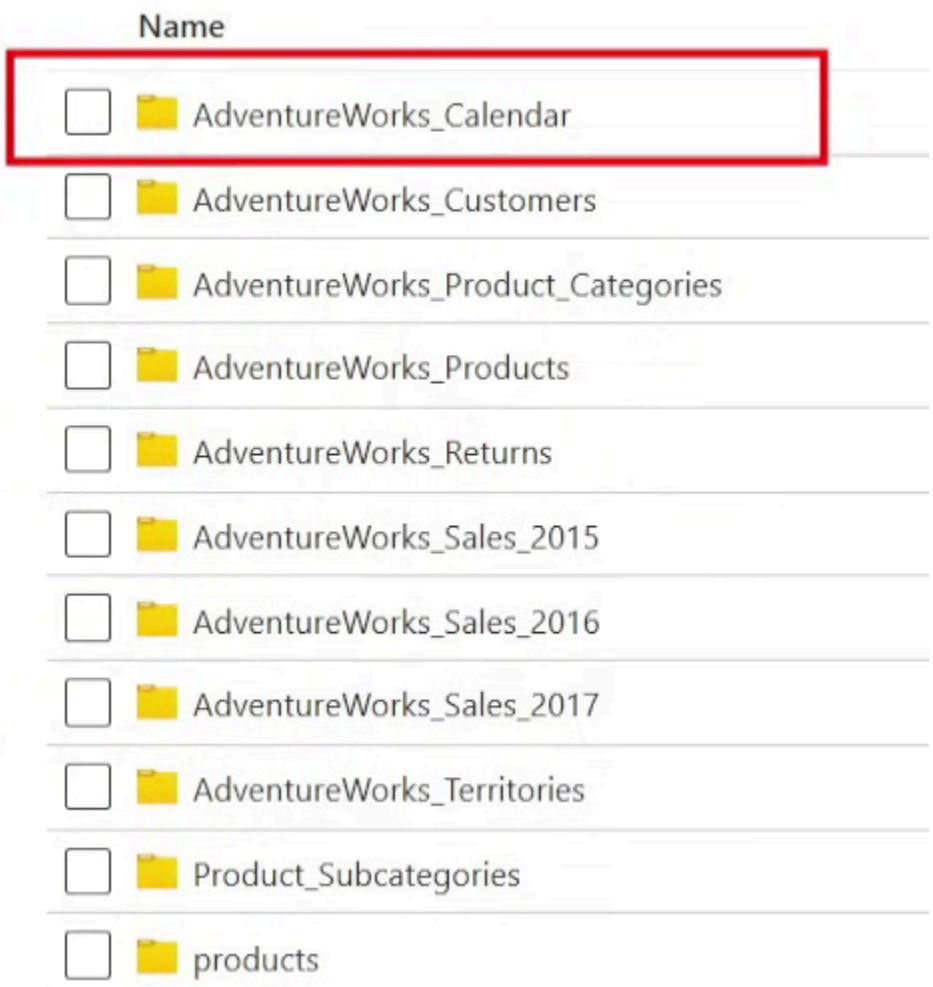
After pasting it remove service credentials starting first line fully

Object ID same as tenant id

Source code is given in the GitHub folder

Now we have called the application to allow databrics to access the files

Step 4 now to load the data from a particular file such as calendar we first need to load it so for loading it we need to write command in python (we will do for each one my one)



Now in this command is true used to basically csv read the data as text so it is used to identify the csv data as csv data

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Silver Layer

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tapanschauhan130@gmail.com
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SILVER LAYER SCRIPT

DATA ACCESS USING APPLICATION

DATA LOADING

4 minutes ago (<1s) 4

```
spark.conf.set(
  "fs.azure.account.auth.type.salesprojectdatalake1.dfs.core.windows.net",
  "OAuth"
)

spark.conf.set(
  "fs.azure.account.oauth.provider.type.salesprojectdatalake1.dfs.core.windows.net",
  "org.apache.hadoop.fs.azurebfs.oauth2.ClientCredsTokenProvider"
)

spark.conf.set(
  "fs.azure.account.oauth2.client.id.salesprojectdatalake1.dfs.core.windows.net",
  "00b3859f-222d-4fbb-9e72-e3fc0aab5483"
```

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tapanschauhan130@gmail.com
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02:18 PM (<1s) 4

```
  "fs.azure.account.oauth2.client.secret.salesprojectdatalake1.dfs.core.windows.net",
  "Oc68Q~wBQiOScHIm7U7Jy1dRYQtkGG9c~sQPJcz-"
)

spark.conf.set(
  "fs.azure.account.oauth2.client.endpoint.salesprojectdatalake1.dfs.core.windows.net",
  "https://login.microsoftonline.com/071d5362-14b7-4f86-9d90-bd0ecbbb9a5f/oauth2/token"
)
```

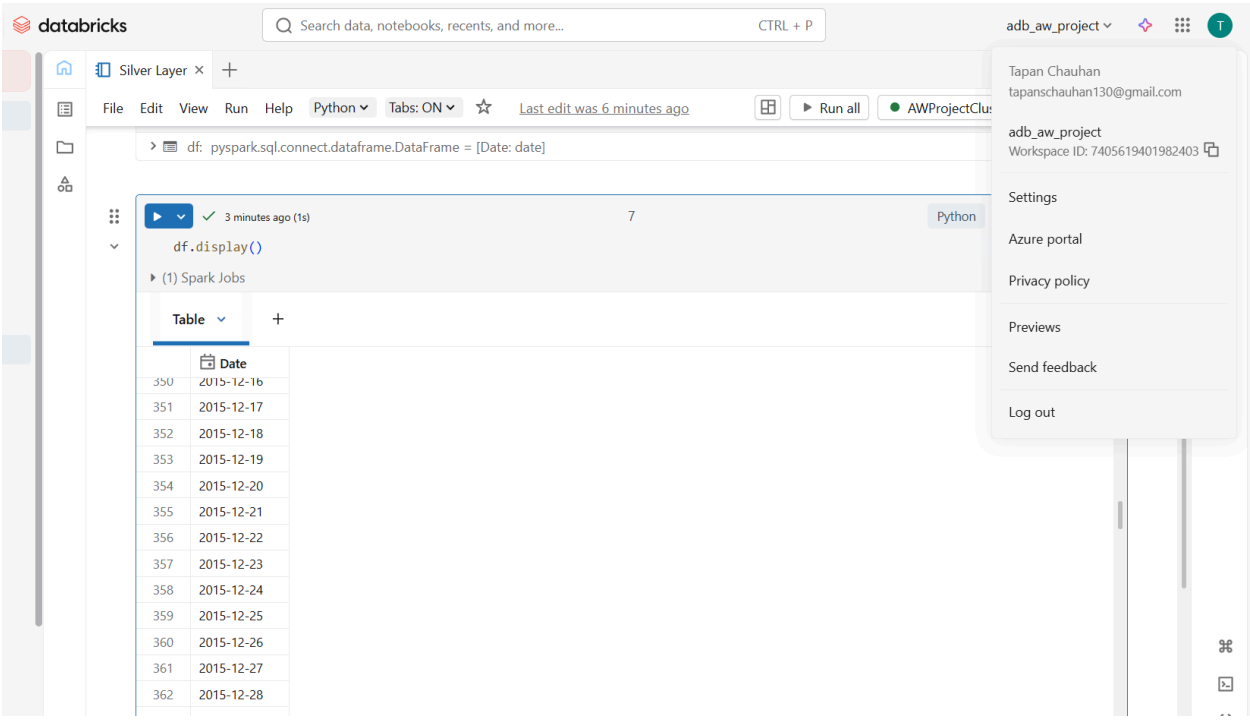
READ Calendar DATA

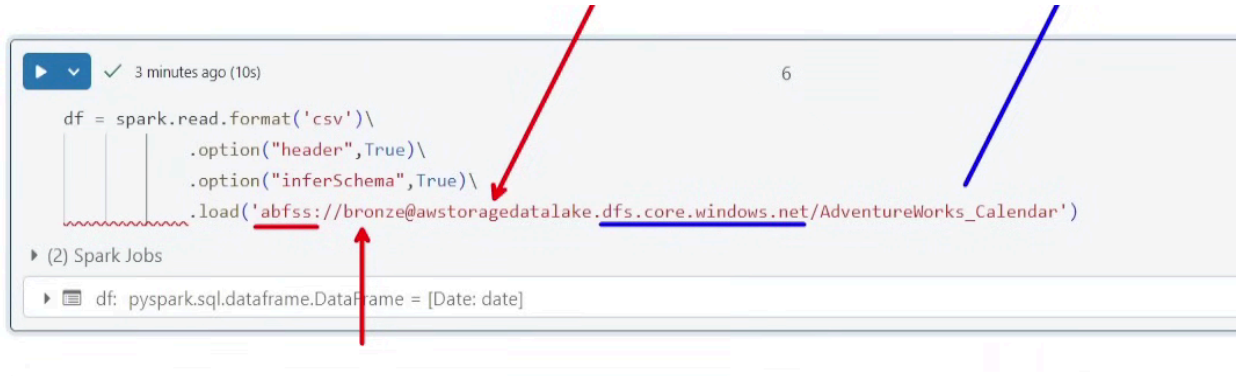
3 minutes ago (13s) 6

```
df = spark.read.format('csv').option("header",True).option("inferschema", True).load('abfss://bronze@salesprojectdatalake1.dfs.core.windows.net/AdventureWorks_Calendar')
```

(2) Spark Jobs

df: pyspark.sql.connect.dataframe.DataFrame = [Date: date]

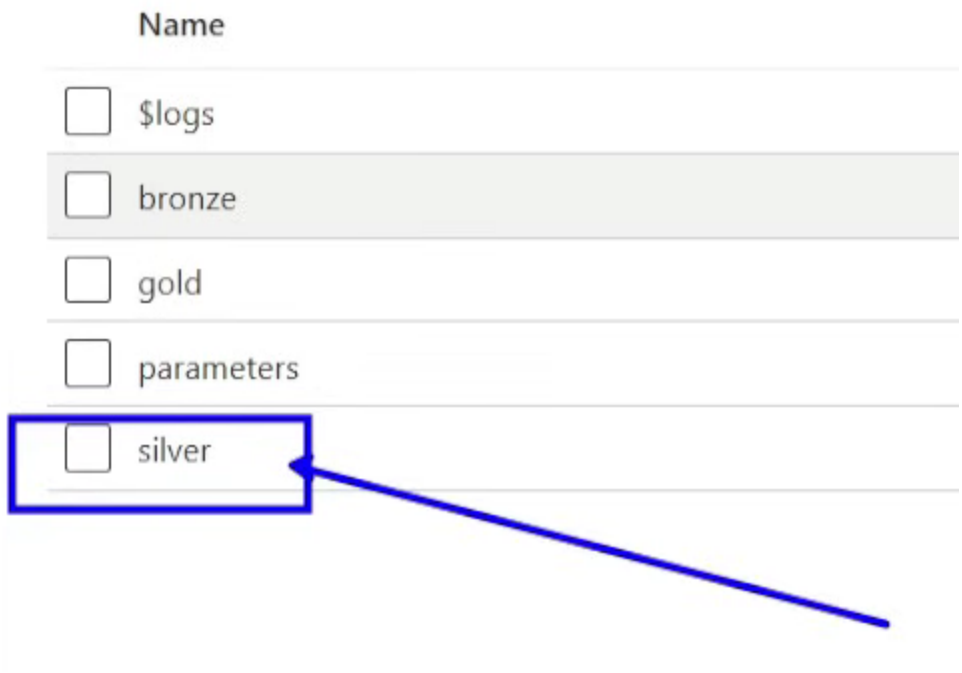




This is asked in interview

Now let's load other data files as well what we can do for this weekend copy paste the code for each but make sure to name the different function as the file name

Pull the library



We will transform the data and load it here

Now we will pull the library

Just now (<1s)

```
from pyspark.sql.functions import*
from pyspark.sql.types import*
```

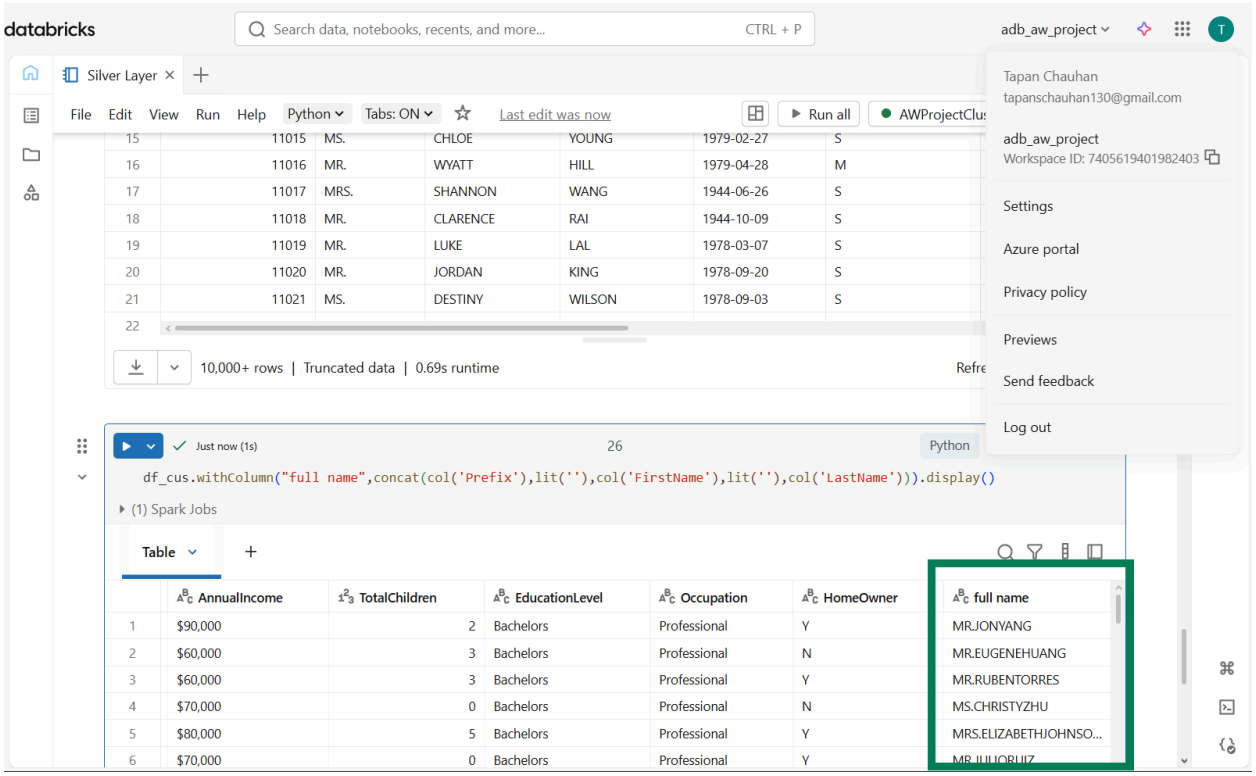
Now with the pySpark writer API We will load the transform data into silver folder
There are four modes available append mode
Overwrite replace
Error mode
Ignore mode

We are using date function

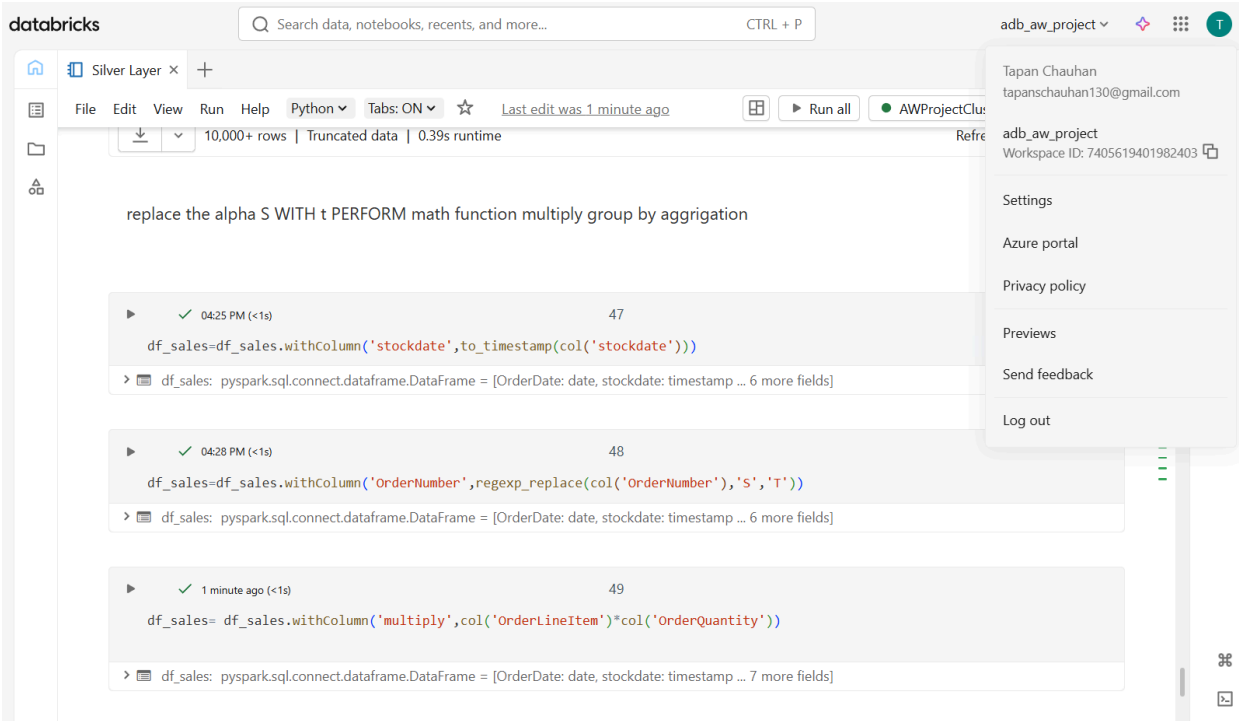
Now we will do customer transformation 3 transformations

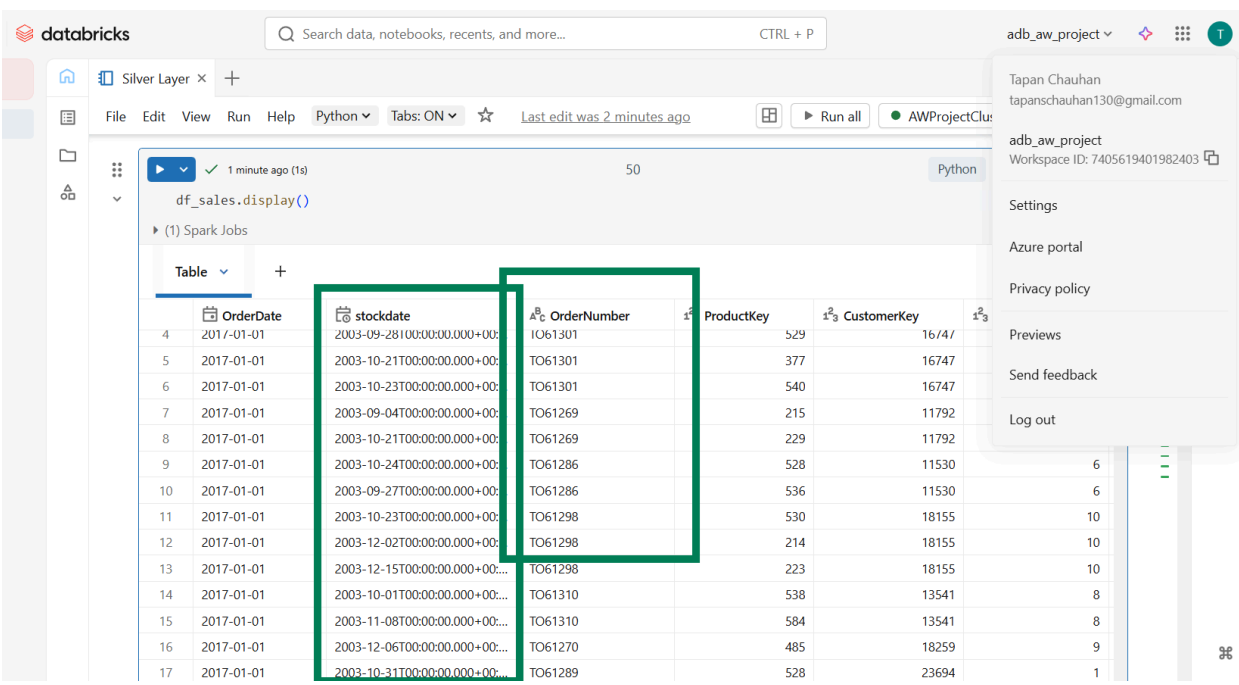
1) String transformation

So here we have 3 columns Mr and Mrs column first name column and last name column and we will combine these column to make a full name

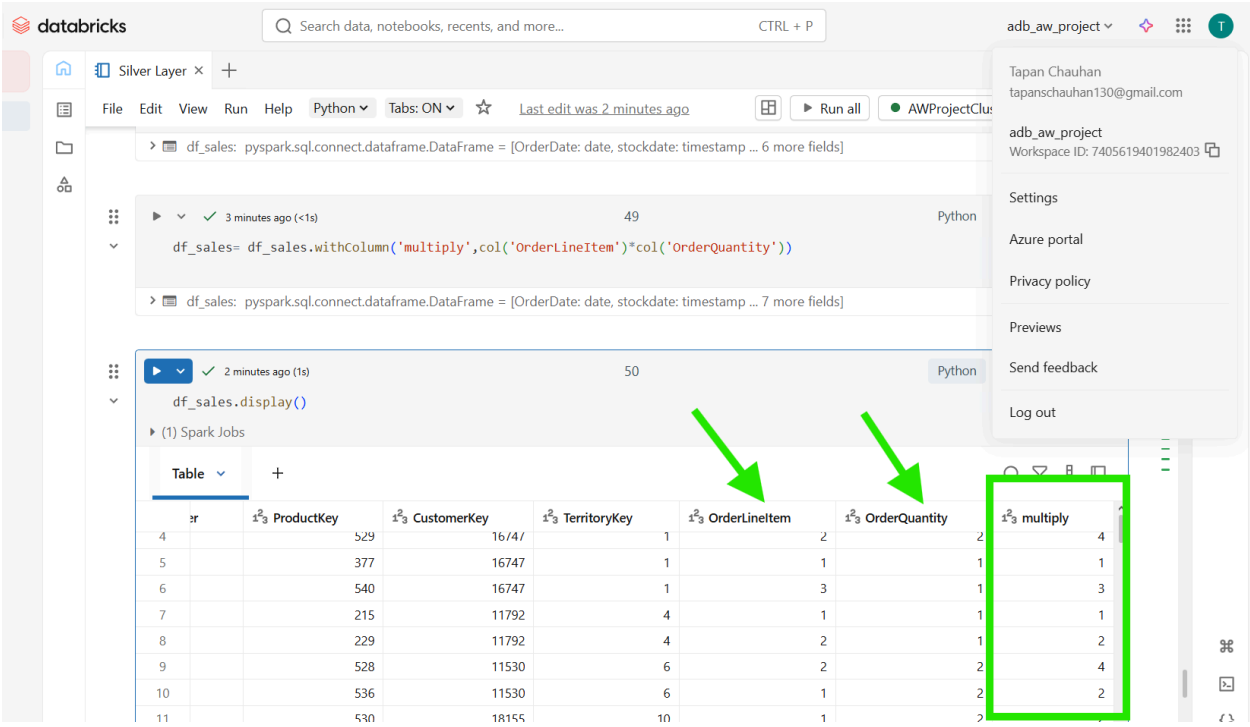


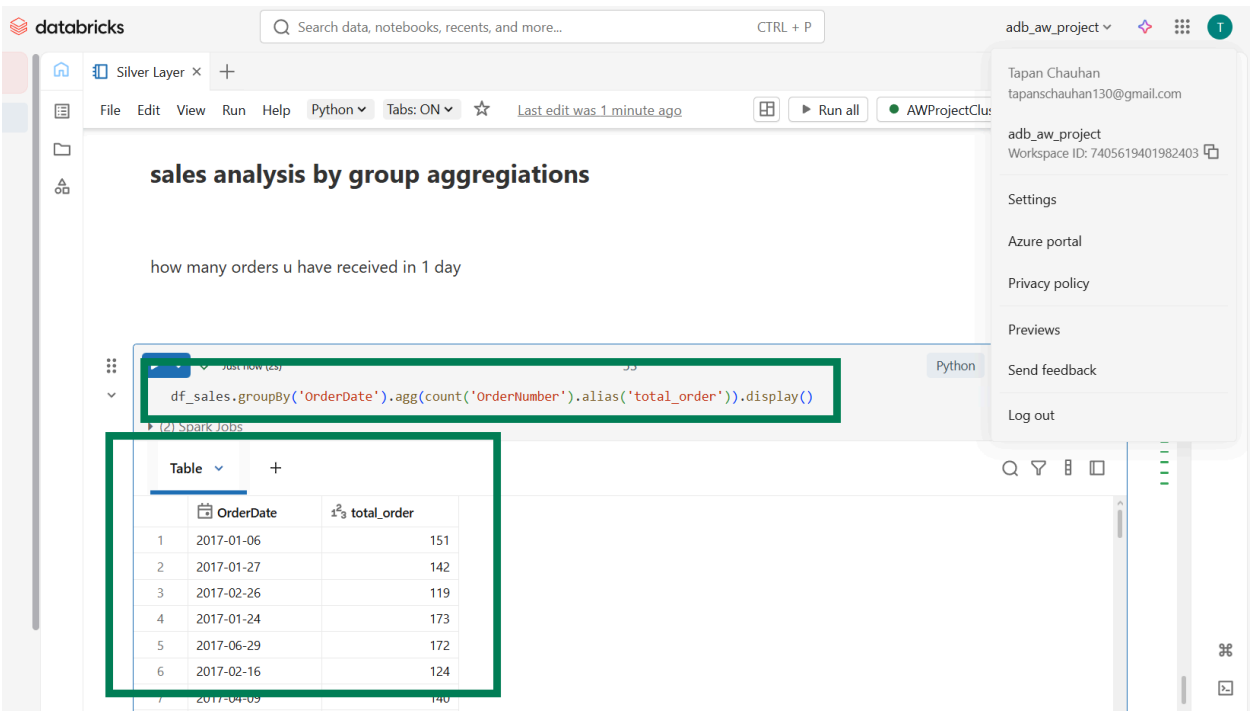
Transformations in sales data



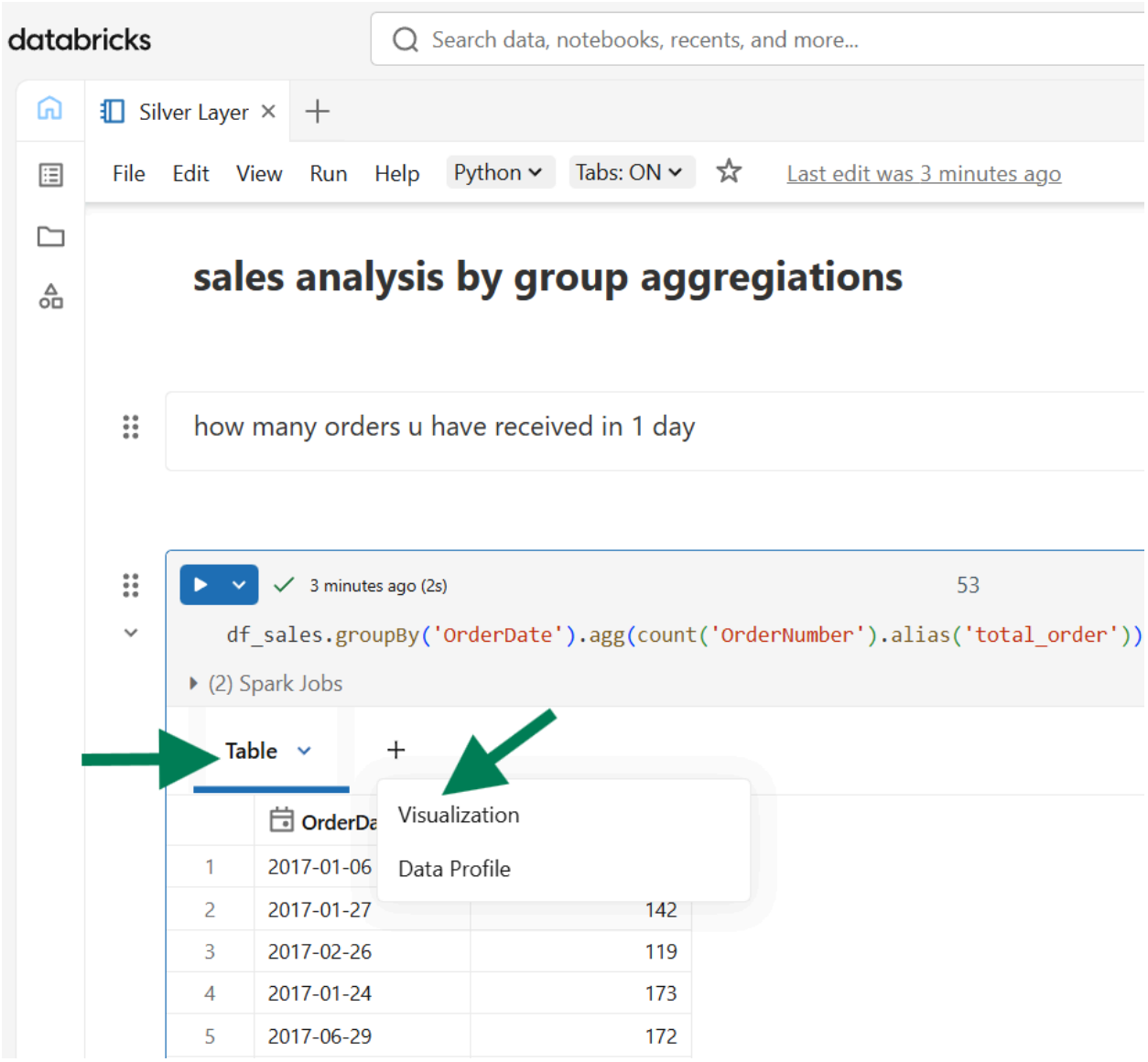


	OrderDate	stockdate	OrderNumber	ProductKey	CustomerKey	
4	2017-01-01	2003-09-28T00:00:00.000+00:	TO61301	529	16747	
5	2017-01-01	2003-10-21T00:00:00.000+00:	TO61301	377	16747	
6	2017-01-01	2003-10-23T00:00:00.000+00:	TO61301	540	16747	
7	2017-01-01	2003-09-04T00:00:00.000+00:	TO61269	215	11792	
8	2017-01-01	2003-10-21T00:00:00.000+00:	TO61269	229	11792	
9	2017-01-01	2003-10-24T00:00:00.000+00:	TO61286	528	11530	6
10	2017-01-01	2003-09-27T00:00:00.000+00:	TO61286	536	11530	6
11	2017-01-01	2003-10-23T00:00:00.000+00:	TO61298	530	18155	10
12	2017-01-01	2003-12-02T00:00:00.000+00:	TO61298	214	18155	10
13	2017-01-01	2003-12-15T00:00:00.000+00:...	TO61298	223	18155	10
14	2017-01-01	2003-10-01T00:00:00.000+00:...	TO61310	538	13541	8
15	2017-01-01	2003-11-08T00:00:00.000+00:...	TO61310	584	13541	8
16	2017-01-01	2003-12-06T00:00:00.000+00:...	TO61270	485	18259	9
17	2017-01-01	2003-10-31T00:00:00.000+00:...	TO61289	528	23694	1





Data visualization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sales analysis by group aggregiations
how many orders u have received in 1 day
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df_sales.groupBy('OrderDate').agg(count('OrderNumber').alias('total_order'))
(2) Spark Jobs
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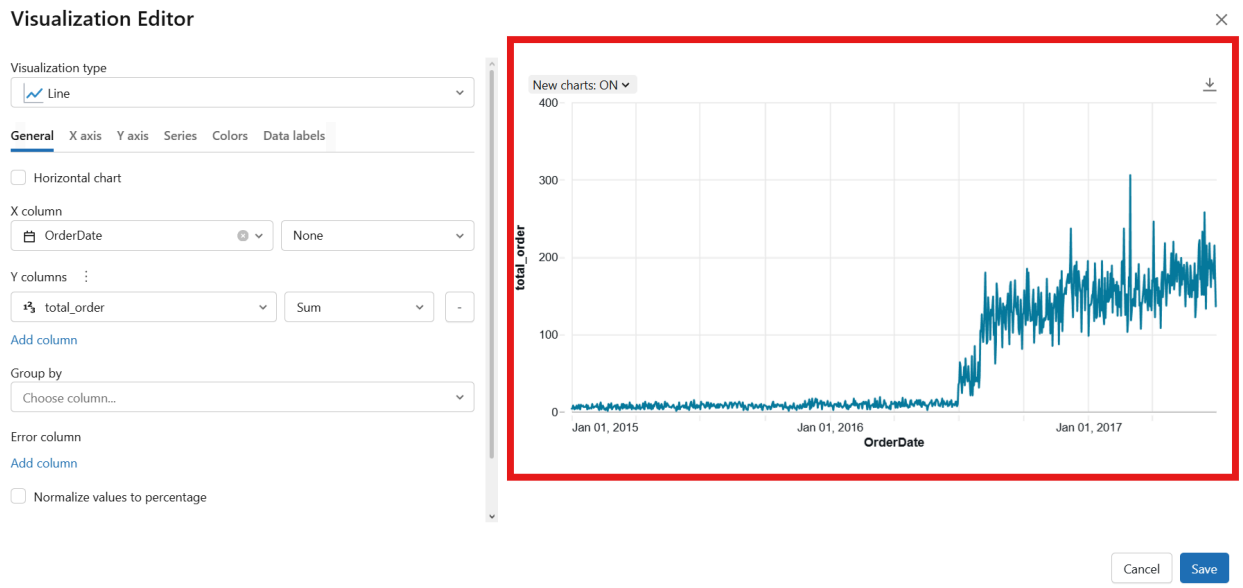
1 2017-01-06
2 2017-01-27 142
3 2017-02-26 119
4 2017-01-24 173
5 2017-06-29 172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General
X axis
Y axis
Series
Colors
Data labels
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Y columns
total_order
Sum
Add column
Group by
Choose column...
Error column
Add column
Normalize values to percentage
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400
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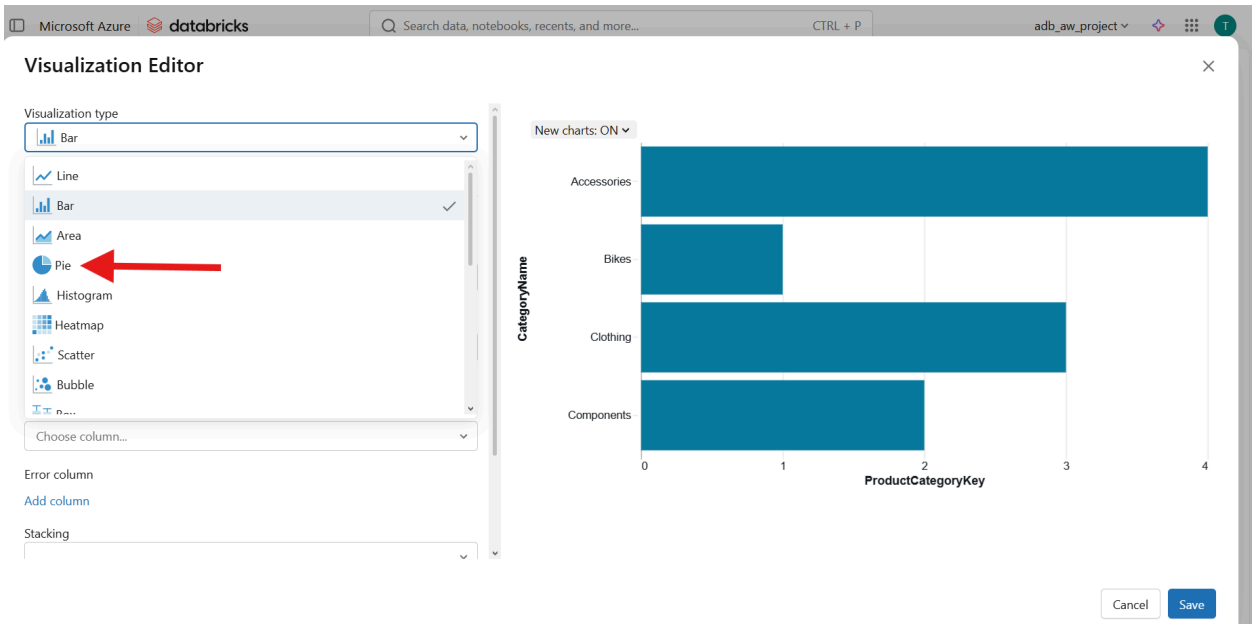
100
0
total_order
Jan 01, 2015
Jan 01, 2016
Jan 01, 2017
OrderDate
Cancel
Save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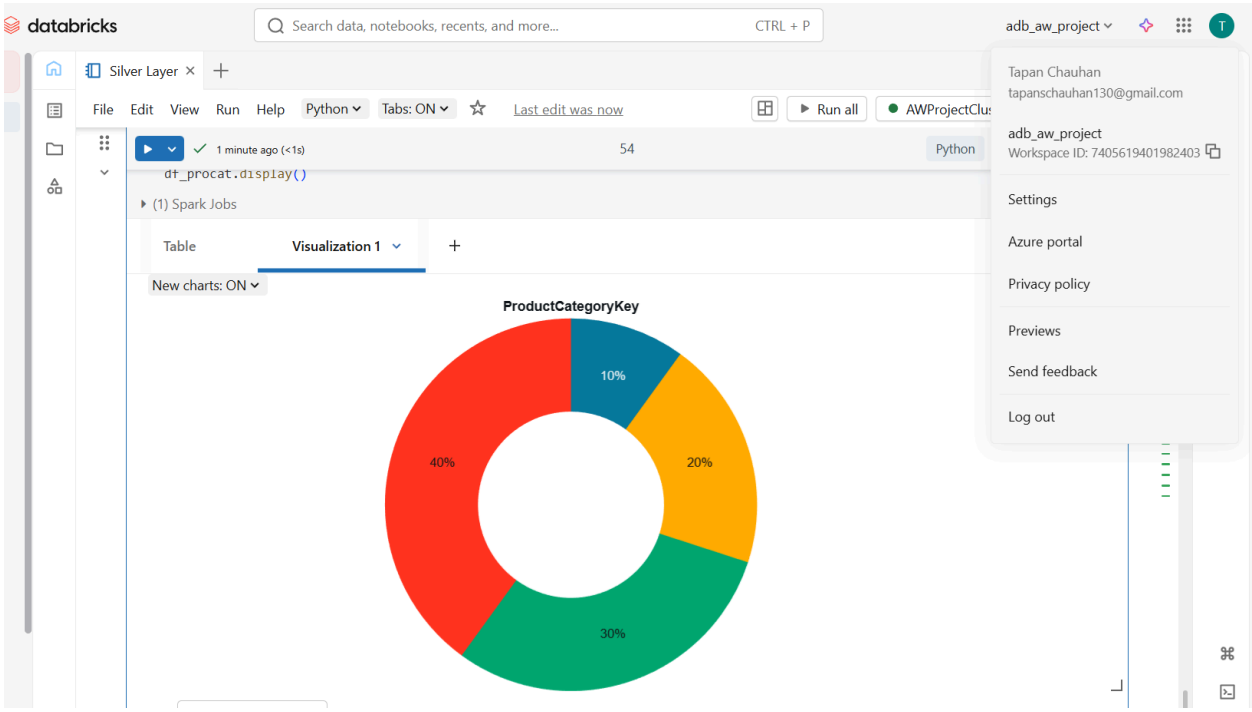
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Python
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(1) Spark Jobs
Table
Visualization 1
New charts: ON
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10%
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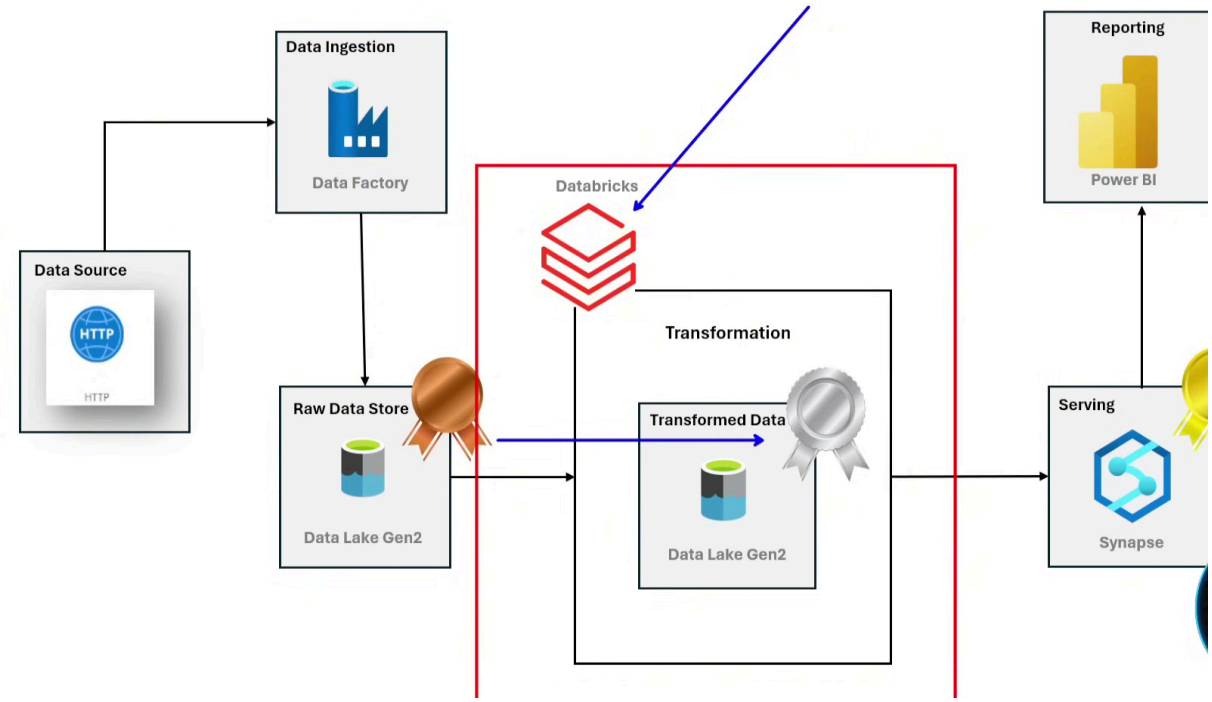
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